

Manifold-Informed Architecture Benchmark

HEART: 2 classes, 13D input

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waverider @ 054030a (-abbrev-re 054030a600978c0e9ffac58faf7157939927d009) | 2026-04-14 22:20:05
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Best: ManifoldModel ($\tau=0.9$) test acc = 0.8382 ± 0.0247
vs Standard: +0.0286 (2.86 pp) inf× fewer params nan× acc/Kparam
Manifold: 13D $\rightarrow$ 9D (30.8% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	HEART
Input dimensionality	13D
Classes	2
Intrinsic dim (d)	9
Variance threshold (τ)	0.9
Epochs	50
Trials	3
Batch size	32
Learning rate	0.001
k neighbors (local PCA)	40

Manifold Discovery

Local PCA over the training set, $k = 40$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 2$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	9.8	0.7	8	11	24.3
0.90	8.3	0.6	7	10	35.9
0.85	7.2	0.6	6	8	44.3
0.80	6.4	0.6	5	7	50.9

Per-Class Intrinsic Dimensionality

Class	Mean d	Std	Min	Max
1	8.6	0.5	8	9
0	8.1	0.7	7	9

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Euclidean KNN (k=7)	0	0.8216 ± 0.0205	—	—
	0	0.8382 ± 0.0247	—	—
ManifoldModel ($\tau=0.9$) $\star$				
Standard (104 $\rightarrow$ 52)	7,022	0.8096 ± 0.0291	0.6827	0.1153
Manifold (2d $\rightarrow$ d, d=9)	146	0.8064 ± 0.0359	0.4295	5.5231
Wide Manifold (d+1=10)	162	0.8227 ± 0.0401	0.4103	5.0785
Manifold+ManifoldAdam (d=9)	146	0.8152 ± 0.0436	0.4036	5.5836
PCA $\rightarrow$ 9D + MLP (2d $\rightarrow$ d)	371	0.8316 ± 0.0331	0.4102	2.2414
Intrinsic Dim (PCA $\rightarrow$ 9D $\rightarrow$ C)	110	0.8271 ± 0.0393	0.4055	7.5191

$\star$ = best performing architecture

Key Findings

- **Best architecture:** ManifoldModel ($\tau=0.9$) — test accuracy 0.8382 ± 0.0247
- **vs Standard:** +0.0286 (2.86 pp) accuracy gain
- **Parameter reduction:** inf $\times$ fewer parameters (0 vs 7,022)
- **Parameter efficiency:** nan $\times$ improvement (nan vs 0.1153 acc/Kparam)
- **Manifold compression:** 13D $\rightarrow$ 9D (30.8% of ambient dimensions carry no structure)

Result Figure

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	054030a (-abbrev-re 054030a600978c0e9ffac58faf7157939927d009)
Commit date	2026-04-14 22:20:05 -0400
Commit message	chore(release): bump version to 0.6.0
Python	3.12.13
TensorFlow	2.21.0
Device	CPU (forced)
Host	Turing
OS	macOS-26.4-arm64-arm-64bit
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
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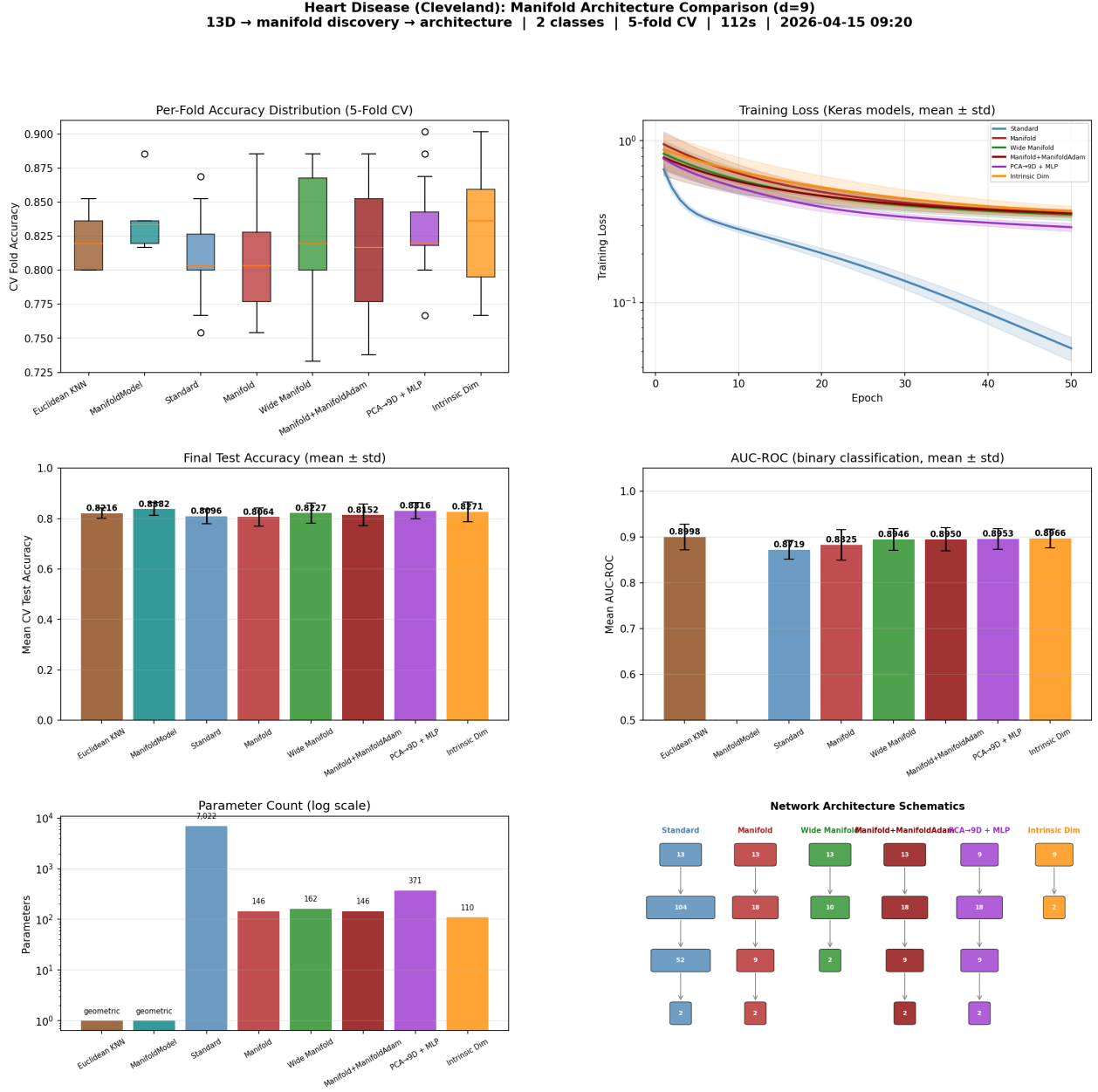


Figure 1: HEART manifold-informed vs. standard architecture comparison. $d = 9$, 50 epochs, 3 trials.